



East West University

Department of Computer Science and Engineering

Semester : spring 2021

Course Title: Electrical Circuits

Course Code: CSE209

Section: 03

Experiment No: 01

Experiment Title: Introduction to Circuit Elements and Variables

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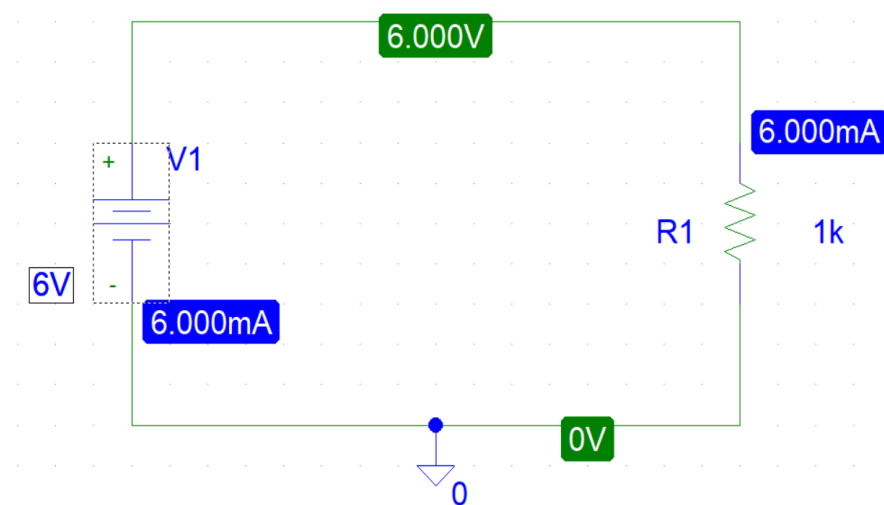
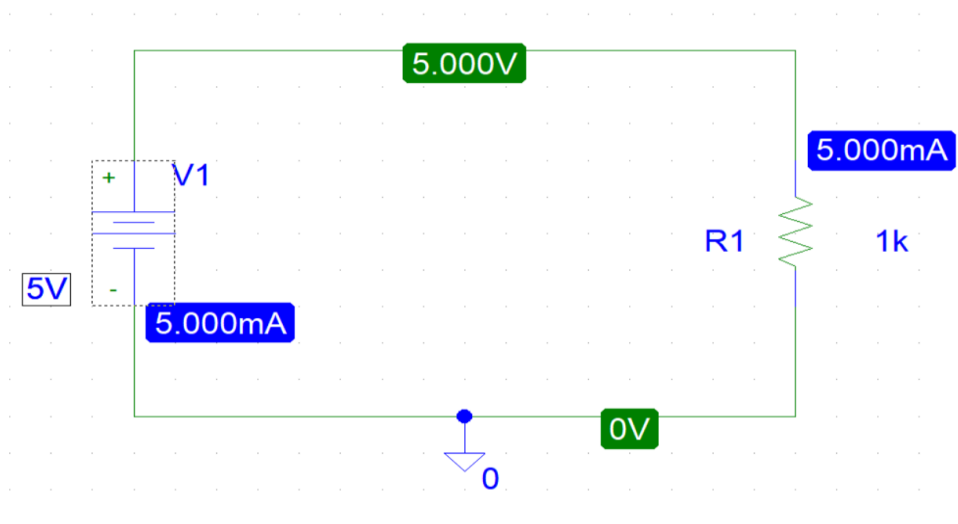
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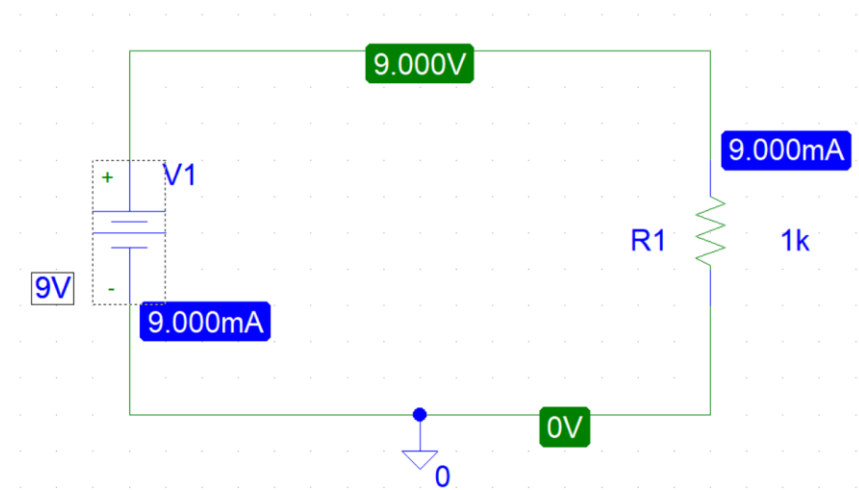
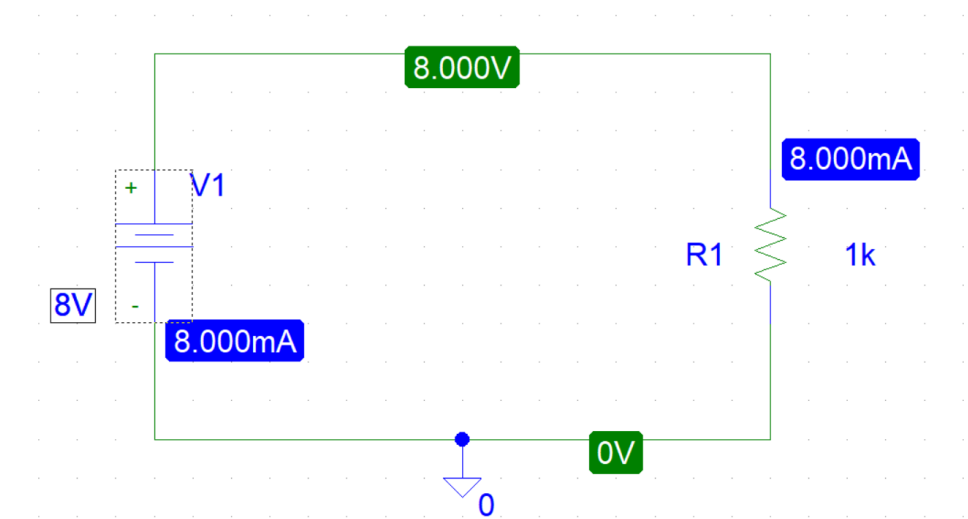
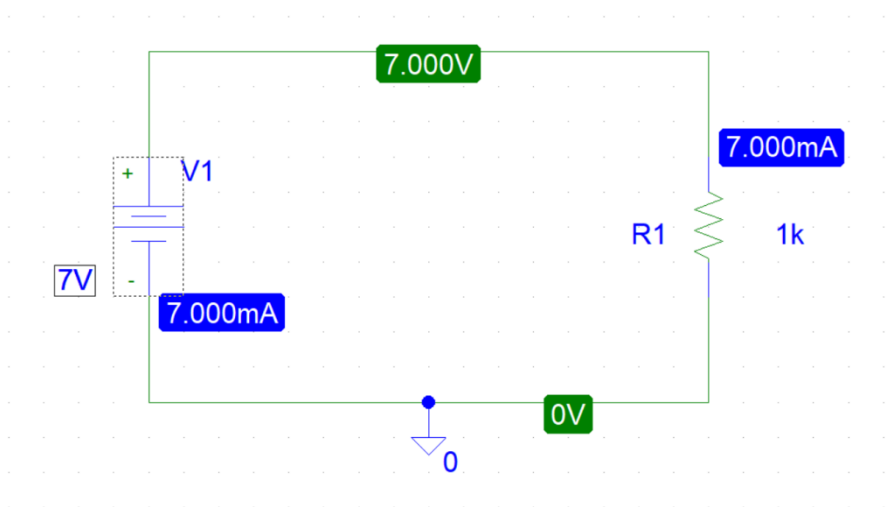
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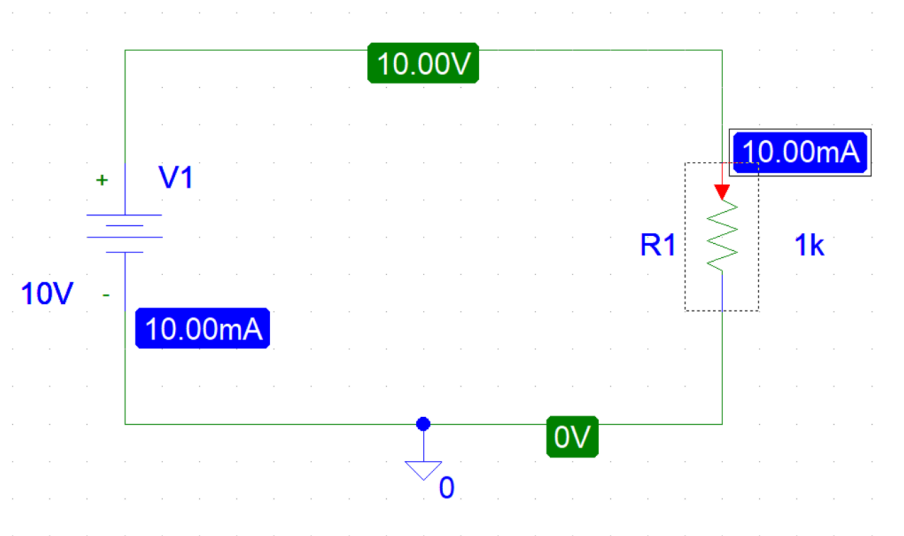
Objectives:

1. To get familiar with circuit variables (voltage and current) and circuit elements (voltage source and resistance).
2. To learn how to measure dc voltage across a circuit element using a voltmeter.
3. To learn how to measure dc current through a circuit element using an ammeter.
4. To learn how to measure resistance of a resistor using a multimeter
5. To verify Ohm's Law.

Circuit Diagram:







Experimental Datasheet:

Observation number	Value of voltage source E (V)	Value of R(Ω)	Measured Value of I(mA)
1	5	1000 Ω	5
2	6	1000 Ω	6
3	7	1000 Ω	7
4	8	1000 Ω	8
5	9	1000 Ω	9
6	10	1000 Ω	10

Answers to the post lab report questions:

1. We have to use ohm's law to calculate the values of I.

From ohm's law, we know,

$$V = IR.$$

Here , given values of E=5V, 6V,7V,8V,9V,10V and R=1000 Ω

When, E=5V

$$I_1 = \frac{V}{R} = \frac{5}{1000} = 5\text{mA}$$

When, E=6V

$$I_2 = \frac{V}{R} = \frac{6}{1000} = 6\text{mA}$$

When, E=7V

$$I_3 = \frac{V}{R} = \frac{7}{1000} = 7\text{mA}$$

When, E=8V

$$I_4 = \frac{V}{R} = \frac{8}{1000} = 8\text{mA}$$

When, E=9V

$$I_5 = \frac{V}{R} = \frac{9}{1000} = 9\text{mA}$$

When, E=10V

$$I_6 = \frac{V}{R} = \frac{10}{1000} = 10\text{mA}$$

As the experiment was done in PSpice, there is no discrepancy.

2. Here $V_1=5\text{V}$, $V_2= 6\text{V}$, $V_3= 7\text{V}$, $V_4= 8\text{V}$, $V_5= 9\text{V}$, $V_6= 10\text{V}$
And $I_1 = 5\text{mA}$, $I_2= 6\text{mA}$, $I_3= 7\text{mA}$, $I_4= 8\text{mA}$, $I_5= 9\text{mA}$, $I_6= 10\text{mA}$
From ohm's law,

$$R_1 = \frac{V}{I} = \frac{5}{.005} = 1000\Omega$$

$$R_2 = \frac{V}{I} = \frac{6}{.006} = 1000\Omega$$

$$R_3 = \frac{V}{I} = \frac{7}{.007} = 1000\Omega$$

$$R_4 = \frac{V}{I} = \frac{8}{.008} = 1000\Omega$$

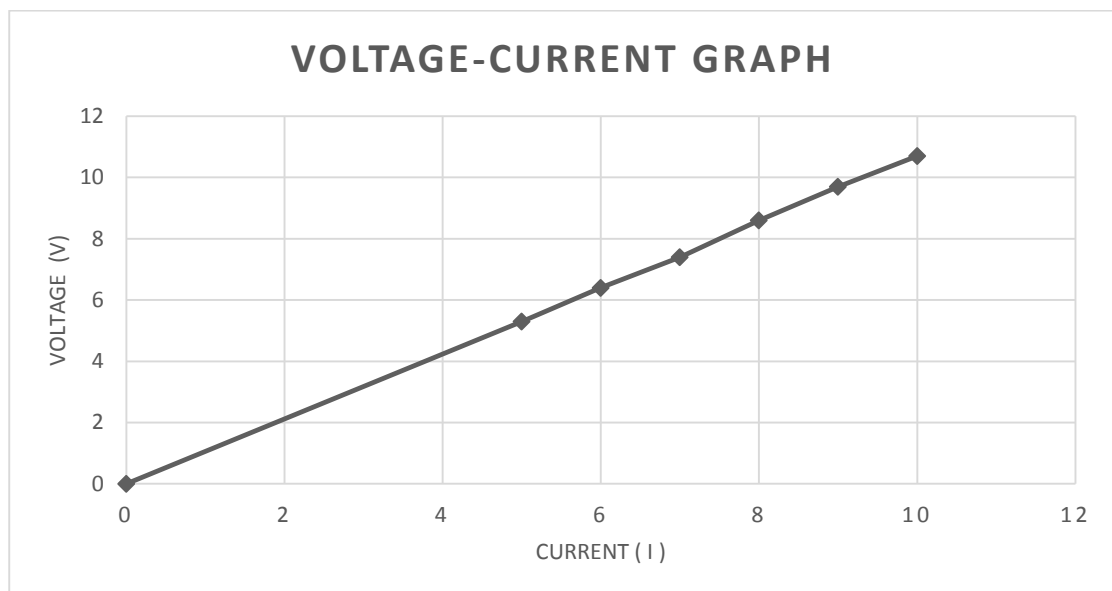
$$R_5 = \frac{V}{I} = \frac{9}{.009} = 1000\Omega$$

$$R_6 = \frac{V}{I} = \frac{10}{.010} = 1000\Omega$$

As the experiment was done in PSpice, there is no discrepancy.

3. As the experiment was done in PSpice, the set values of E and the value of V are same. Thus , there is no discrepancy.

4.



5.The experiment was done in Pspice.So the multi-range meter was not used in the experiment

Conclusion:

In this experiment we learnt how to construct a circuit in PSpice & how to simulate it. As the experiment was done in PSpice, the values of V & R was the same as given value. And there was no discrepancy.

